

# Neural Network Optimization as a Constrained Nonlinear Program

Arindam Paul

Numerical Optimization Project

Albert-Ludwigs-Universität Freiburg, Summer Term 2026

**Abstract**—I study small neural-network regression written as a constrained nonlinear program (NLP): the network weights are the decision variables, the mean-squared error is the objective, and two requirements are imposed as hard constraints, a bound on the network’s input Lipschitz behaviour and a Euclidean norm ball on the weights. The true Lipschitz constant is not directly convenient for the solver, so I enforce a computable certificate  $U(w) = \|w_1\|_2 \|w_2\|_2$ , a guaranteed upper bound whose constraint  $U(w) \leq L$  is a conservative sufficient condition for an  $L$ -Lipschitz map. I build the program in CasADi and solve it with the interior-point solver IPOPT, comparing against matched soft-penalty Adam training on the same requirements. IPOPT satisfies both constraints to solver tolerance in every trial, while weight decay leaves the certificate uncontrolled; a converged single-constraint penalty shows a feasibility transition near the associated multiplier. Enlarging the network in width or depth yields no accuracy gain but makes the interior-point solve markedly more expensive.

**Index Terms**—constrained optimization, interior-point methods, neural networks, Lipschitz bound, IPOPT, CasADi.

## I. INTRODUCTION

Suppose a regression model must satisfy a guaranteed bound on how fast its output can change when its input is perturbed, a Lipschitz bound used to certify robustness or smoothness. Soft Lipschitz regularization adds a penalty that favours small layer norms [5], and spectral normalization rescales each layer so that its operator norm is controlled directly [6]. This project studies a different, explicit small-scale route: a conservative Lipschitz certificate is imposed as a hard nonlinear constraint, and the resulting program is solved with the interior-point solver IPOPT [2] through the symbolic framework CasADi [1]. Training is then literally a constrained NLP: the weights are the decision variables, the fitting error is the objective, and the requirements are constraints the solver enforces rather than merely favours.

The two requirements, a Lipschitz bound  $L$  and a parameter-norm bound  $B$ , are direct modelling choices. The hard formulation removes the penalty coefficient  $\rho$  of a soft method, but it does not remove the responsibility to choose meaningful limits  $L$  and  $B$ . A useful by-product for a numerical-optimization reader is that IPOPT returns a Lagrange multiplier for each active constraint, which under standard regularity assumptions gives a local sensitivity of the fitting error to the corresponding bound.

**Contributions.** (i) A hard-constrained NLP formulation of training in which a computable Lipschitz certificate and a

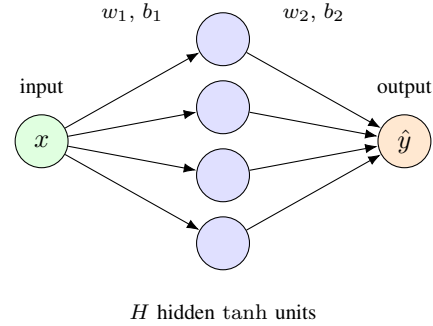


Fig. 1. The student network of (1): one scalar input,  $H$  hidden tanh units, one scalar output. All weights and biases are the decision variables of the NLP.

norm ball are enforced directly (Section II). (ii) A like-for-like comparison of the hard-constrained solve against matched soft penalties (Section III). (iii) A synthetic and a simulated-pendulum case study of the resulting models (Section IV). (iv) A study of the interior-point cost as the network grows in *width and depth* (Sections IV and V).

## II. PROBLEM FORMULATION

### A. Network and objective

The model is a one-hidden-layer network with scalar input and output and  $H$  hidden tanh units (Fig. 1),

$$\hat{y}(x; w) = w_2^\top \tanh(w_1 x + b_1) + b_2, \quad (1)$$

with  $w_1, b_1, w_2 \in \mathbb{R}^H$ ,  $b_2 \in \mathbb{R}$ , stacked into one decision vector  $w = (w_1, b_1, w_2, b_2) \in \mathbb{R}^n$ ,  $n = 3H + 1$  (for the main study  $H = 8$ ,  $n = 25$ ). For  $N$  training pairs  $(x_i, y_i)$  the objective is the mean-squared error

$$f(w) = \frac{1}{N} \sum_{i=1}^N (\hat{y}(x_i; w) - y_i)^2. \quad (2)$$

### B. The desired constrained problem

The modelling goal is to fit the data while guaranteeing that the network does not react too fast to input changes. The relevant quantity is the true input Lipschitz constant of the model, written with its explicit dependence on the weights  $w$ ,

$$\text{Lip}_x(\hat{y}(\cdot; w)) = \sup_{x \neq x'} \frac{|\hat{y}(x; w) - \hat{y}(x'; w)|}{|x - x'|}. \quad (3)$$

The ideal training problem is then

$$\begin{aligned} & \underset{w}{\text{minimize}} && f(w) \\ & \text{subject to} && \text{Lip}_x(\hat{y}(\cdot; w)) \leq L, \\ & && \|w\|_2^2 \leq B^2, \end{aligned} \quad (4)$$

where  $f(w)$  is the training error (2), the first constraint is the desired Lipschitz requirement with bound  $L$ , and the second is an explicit parameter-norm ball of radius  $B$ . The supremum in (3) is not directly convenient for a numerical NLP solver, so it is replaced next by a conservative, computable sufficient condition.

### C. Derivation of the tractable certificate

Because (1) is differentiable, its Lipschitz constant is the supremum of  $|\text{d}\hat{y}/\text{d}x|$ . Differentiating,

$$\frac{\text{d}\hat{y}}{\text{d}x} = w_2^\top \text{diag}(1 - \tanh^2(w_1 x + b_1)) w_1. \quad (5)$$

Each diagonal entry lies in  $(0, 1]$ , so the diagonal matrix has operator norm at most one, and Cauchy–Schwarz gives

$$\text{Lip}_x(\hat{y}(\cdot; w)) \leq \|w_1\|_2 \|w_2\|_2 =: U(w). \quad (6)$$

The certificate  $U(w)$  is computable and is an *upper bound*: it is not equal to the true Lipschitz constant in general, and  $U(w) \leq L$  is a conservative sufficient condition for the true requirement  $\text{Lip}_x(\hat{y}(\cdot; w)) \leq L$ . I impose it in the squared form

$$\|w_1\|_2^2 \|w_2\|_2^2 \leq L^2, \quad (7)$$

a degree-four (biquadratic) polynomial in the weights: smooth, but nonconvex. Replacing the true Lipschitz constraint by (7) gives the tractable NLP actually solved,

$$\begin{aligned} & \underset{w}{\text{minimize}} && f(w) \\ & \text{subject to} && \|w_1\|_2^2 \|w_2\|_2^2 \leq L^2, \\ & && \|w\|_2^2 \leq B^2. \end{aligned} \quad (8)$$

## III. NUMERICAL OPTIMIZATION

### A. Hard-constrained solution

The objective and constraints of (8) are assembled symbolically in CasADi [1], which supplies exact first and second derivatives, and the smooth nonconvex program is solved by the interior-point method IPOPT [2]. The hard solves run to a KKT tolerance of  $10^{-8}$ ; here feasibility means the constraints are satisfied to solver tolerance, not that a global optimum has been certified.

### B. Soft-penalty baseline and protocol

The baselines replace the hard constraints by penalties. *Plain Adam* [3] minimizes only the MSE; *AdamW* [4] adds decoupled weight decay with coefficient  $w_d$ , a soft stand-in for the norm ball; *penalty-Adam* adds both requirements as a hinge penalty

$$\min_w f(w) + \rho [\max(0, \|w_1\|_2^2 \|w_2\|_2^2 - L^2) + \max(0, \|w\|_2^2 - B^2)], \quad (9)$$

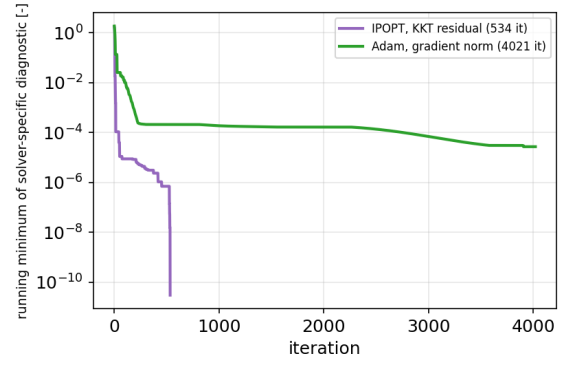


Fig. 2. Running minimum of solver-specific optimality diagnostics versus iteration. IPOPT shows  $\max(\text{inf\_pr}, \text{inf\_du})$  for the hard-constrained solve; plain Adam shows the infinity norm of its gradient. The quantities are different and are compared only qualitatively.

minimized with Adam, where  $\rho \geq 0$ : too small and the requirements are violated, too large and the nonsmooth term stiffens the first-order optimization.

*Stopping and initialization.* Every gradient run stops at a loss plateau (successive change below  $10^{-9}$ ) with a cap of 40 000 iterations; reaching the cap is reported as non-converged. The hard IPOPT solves use tolerance  $10^{-8}$ , and the exact-penalty sweep of Section IV-A uses IPOPT tolerance  $10^{-10}$  with acceptable tolerance  $10^{-9}$ . Within any matched comparison all methods start from the same initialization; the initialization changes with the seed across repeated trials.

## IV. CASE STUDIES

Two regression tasks illustrate the formulation: a synthetic teacher–student problem and a simulated damped pendulum.

### A. Synthetic case study

The synthetic tasks draw inputs  $x_i \sim \text{Uniform}[-3, 3]$  and targets  $y_i = y_T(x_i; w^*) + \varepsilon_i$  with  $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ , where the teacher  $y_T$  is a seeded one-hidden-layer tanh network of six units and the student has  $H$  units. Changing the seed changes the teacher weights, the input sample, the noise and the initialization together, so a set of seeds gives independently seeded teacher–student trials rather than repeated noise on one fixed target.

1) *Optimizer diagnostic:* Fig. 2 plots, against iteration, a running-minimum optimality diagnostic for the two solver families. For IPOPT on the hard-constrained NLP it is the running minimum up to iteration  $k$  of  $\max(\text{inf\_pr}, \text{inf\_du})$ , IPOPT’s primal and dual infeasibility measures; for plain Adam it is the running minimum of  $\|\nabla f(w_j)\|_\infty$ , the infinity norm of the plain-Adam objective gradient at iteration  $j$ . These are different, solver-specific quantities, and plain Adam here minimizes only the MSE rather than solving the hard-constrained NLP, so their absolute vertical values are not directly comparable; the plot shows only the qualitative evolution of each method’s own diagnostic.

TABLE I  
CERTIFICATE  $U(w) = \|w_1\| \|w_2\|$  OVER 15 INDEPENDENTLY SEEDED TRIALS (MEDIAN [min, max]) AND THE NUMBER THAT VIOLATE  $L = 4$ . IPOPT STAYS ON THE BOUND TO SOLVER TOLERANCE; THE ADAM BASELINES ARE UNCONTROLLED AND WORSEN WITH NOISE.

| $\sigma$ | Method                    | $U(w)$ (median [min, max]) | Viol. of 4 |
|----------|---------------------------|----------------------------|------------|
| 0.1      | IPOPT (hard)              | 4.00 [4.00, 4.00]          | 0/15       |
| 0.1      | plain Adam                | 5.83 [1.27, 24.0]          | 10/15      |
| 0.1      | AdamW ( $w_d = 0.01$ )    | 3.19 [0.91, 11.1]          | 5/15       |
| 0.3      | IPOPT (hard)              | 4.00 [4.00, 4.00]          | 0/15       |
| 0.3      | plain Adam                | 63.4 [6.7, 399]            | 15/15      |
| 0.3      | AdamW ( $w_d = 10^{-4}$ ) | 44.2 [6.7, 153]            | 15/15      |

2) *Certificate control*: Can a method *set* the certificate rather than merely influence it? I fit the synthetic target with  $H = 8$ ,  $L = 4$ ,  $B = 6$ , on 60 training and 40 test points, at two noise levels  $\sigma \in \{0.1, 0.3\}$ , over 15 independently seeded trials. Table I reports the certificate  $U(w) = \|w_1\| \|w_2\|$  of the hard solve and of the Adam baselines. IPOPT satisfies  $U(w) \leq 4$  in every trial, with the certificate active near the boundary to solver tolerance (median 4.00 at both noise levels, zero violations). Plain Adam does not control it, and worsens with noise; AdamW improves matters but still violates the bound in a third of the low-noise trials and in all high-noise trials. Its weight-decay coefficient  $w_d$  was chosen on a pilot seed (seed 0) by test error, to give the baseline a strong setting; this is a transparent modelling choice, not a validation-tuned result.

3) *Exact-penalty analysis*: This experiment specializes the soft penalty (9) to the single certificate constraint and solves it to convergence with an IPOPT epigraph reformulation, which isolates the formulation effect from a fixed-budget Adam run. On a fixed teacher (seed 2,  $\sigma = 0.2$ , 40 training / 40 validation / 40 test points, only the certificate constraint included) the hard reference sits on the bound at  $U(w) = 4.00$  with KKT multiplier  $\lambda^* = 1.6 \times 10^{-4}$ , training MSE 0.033 and test MSE 0.066. Fig. 3 shows a feasibility transition close to this multiplier: below  $\lambda^*$  the certificate exceeds  $L$  (about 7.9, 6.6, 5.8), and the first sampled value above  $\lambda^*$ ,  $\rho = 1.7 \times 10^{-4}$ , already sits on the bound, as do the subsequent sampled values. Because the problem is nonconvex, exact-penalty theory [7] guarantees only local exactness once  $\rho$  exceeds the relevant multiplier; I read the plot as a feasibility transition near the hard solution’s multiplier, not as proof that every larger  $\rho$  returns the identical parameter vector.

4) *Width scaling*: Bigger networks do not fit this target better, but the hard solve costs steeply more. The width experiment uses  $\sigma = 0.05$ , seed 0,  $L = 4$ ,  $B = 6$ , and 60 training / 40 test points, and compares IPOPT against penalty-Adam on the same two requirements. Growing  $H$  from 4 to 256 ( $n$  from 13 to 769; Fig. 4) leaves the test error near 0.003 for both methods, so the added capacity buys no accuracy. IPOPT satisfies the certificate bound in every run, with the certificate generally near the boundary ( $U(w) \approx 4.00$ ); its solve time rises from about 0.14s to about 453s, while penalty-Adam takes about 13.2s. On this

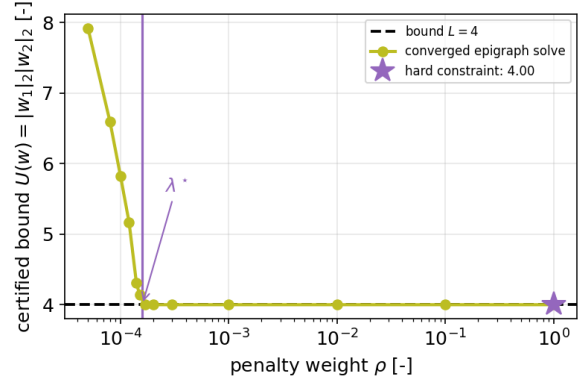


Fig. 3. Converged exact-penalty sweep for the certificate constraint. The certified bound  $U(w)$  of the converged epigraph solve exceeds  $L = 4$  for  $\rho < \lambda^*$ ; the first sampled value above  $\lambda^*$ , and subsequent sampled values, lie on the boundary, matching the hard-constrained IPOPT reference.

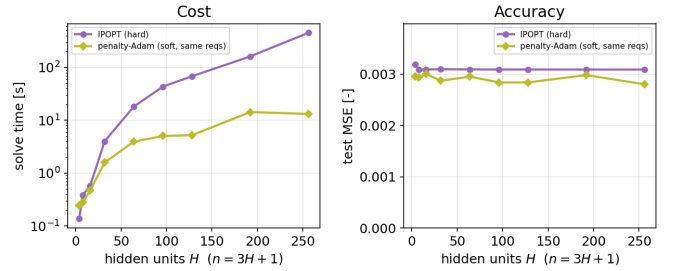


Fig. 4. Width scaling from  $H = 4$  to 256: IPOPT cost grows steeply, while test MSE remains nearly flat for both methods.

implementation and the measured range ( $n \geq 49$ ) the interior-point time grows empirically about as  $n^{2.3}$ ; this is a fit to the measured points of one run, not a universal complexity law, and wall-clock timings are not expected to reproduce exactly. The mechanism is that each interior-point iteration solves a KKT linear system whose cost grows with the number of decision variables.

#### B. Simulated damped-pendulum case study

As a small dynamics-flavoured task I fit the *simulated* free response of a linearized damped pendulum,

$$\theta(t) = \theta_0 e^{-\zeta \omega t} \left[ \cos(\omega_d t) + \frac{\zeta \omega}{\omega_d} \sin(\omega_d t) \right], \quad (10)$$

$$\omega_d = \omega \sqrt{1 - \zeta^2}.$$

with  $\theta_0 = 1$  rad,  $\omega = 2$  rad/s,  $\zeta = 0.15$  and  $t \in [0, 6]$  s; this is a simulated response, not a measurement of a physical rig. Observations are  $t_i \sim \text{Uniform}[0, 6]$ ,  $y_i = \theta(t_i) + \varepsilon_i$ ,  $\varepsilon_i \sim \mathcal{N}(0, 0.15^2)$ , split into 60 training / 60 validation / 60 test points;  $\rho$  for the soft method is chosen on the validation split and the reported error is on the test split. The bound  $L = 4$  rad/s is a fixed demonstration requirement on the learned model’s rate  $|\dot{\theta}/dt|$ , not a value inferred from the physics.

TABLE II

PENDULUM: THE HARD SOLVE VERSUS MATCHED PENALTY-ADAM. ONLY IPOPT CONVERGES; PENALTY-ADAM REACHES THE ITERATION CAP WITH A NUMERICALLY FEASIBLE BUT NON-CONVERGED ITERATE.

| Method              | $U(w)$ | $\ w\ $ | Test MSE | Converged?        |
|---------------------|--------|---------|----------|-------------------|
| IPOPT (hard)        | 4.00   | 6.00    | 0.0232   | yes ( $10^{-8}$ ) |
| penalty-Adam (soft) | 3.93   | 5.94    | 0.0243   | no (cap)          |

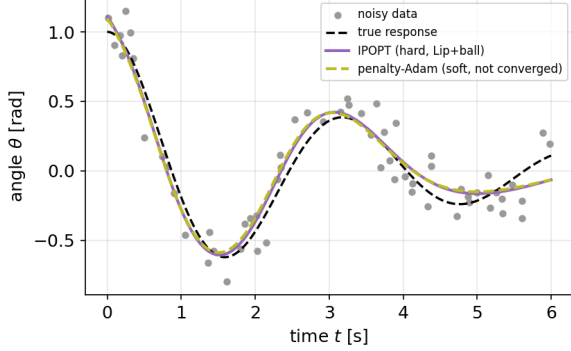


Fig. 5. Damped-pendulum fits: noisy samples (grey), the simulated true response (dashed) and the two matched models. IPOPT certifies  $|\dot{\theta}| \leq 4 \text{ rad s}^{-1}$ ; penalty-Adam fits similarly but did not converge.

Table II and Fig. 5 compare the hard solve with matched penalty-Adam. IPOPT converges in 103 iterations (0.09 s), satisfying both constraints ( $U(w) = 4.00$ ,  $\|w\| = 6.00$ ) at test MSE 0.0232. penalty-Adam at  $\rho = 0.01$  reaches the 40 000-iteration cap without meeting its convergence criterion; its final iterate is numerically feasible ( $U(w) = 3.93$ ,  $\|w\| = 5.94$ ) at test MSE 0.0243. The test errors are close: the difference is direct constraint enforcement and reliable convergence, not predictive accuracy.

## V. EXTENSION TO DEEPER NEURAL NETWORKS

The one-hidden-layer exposition used vectors because the input and output are scalar. A network with  $D$  hidden layers of width  $H$  is defined by

$$\begin{aligned} h_1 &= \tanh(W_1 x + b_1), \\ h_l &= \tanh(W_l h_{l-1} + b_l), \quad l = 2, \dots, D, \\ \hat{y}(x; w) &= W_{D+1} h_D + b_{D+1}, \end{aligned} \quad (11)$$

with  $W_1 \in \mathbb{R}^{H \times 1}$  a column,  $W_l \in \mathbb{R}^{H \times H}$  genuine square matrices for  $l = 2, \dots, D$ , and  $W_{D+1} \in \mathbb{R}^{1 \times H}$  a row. Although the overall network maps  $\mathbb{R}$  to  $\mathbb{R}$ , every hidden state  $h_l$  lies in  $\mathbb{R}^H$ , so the internal maps  $W_l : \mathbb{R}^H \rightarrow \mathbb{R}^H$  are genuinely multidimensional. Their relevant size is the spectral norm

$$\|W\|_2 := \sup_{v \neq 0} \frac{\|Wv\|_2}{\|v\|_2}, \quad (12)$$

the induced Euclidean norm, which is the Lipschitz constant of the affine map  $z \mapsto Wz + b$ . Since  $\tanh$  is 1-Lipschitz and

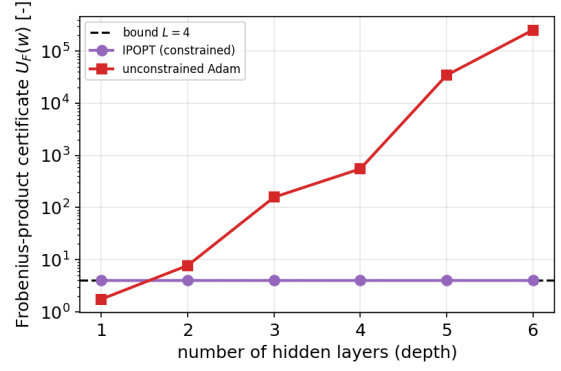


Fig. 6. Frobenius-product certificate  $U_F(w) = \prod_l \|W_l\|_F$  versus depth (log scale). The constraint keeps it near  $L = 4$  while satisfying the bound at every depth; unconstrained Adam grows past  $2.5 \times 10^5$  at six hidden layers.

Lipschitz constants multiply under composition, the natural multilayer bound is the product of the layer spectral norms,

$$\text{Lip}_x(\hat{y}(\cdot; w)) \leq \prod_{l=1}^{D+1} \|W_l\|_2 \leq \prod_{l=1}^{D+1} \|W_l\|_F =: U_F(w). \quad (13)$$

The second inequality uses the Frobenius norm

$$\|W\|_F^2 = \sum_{i,j} W_{ij}^2, \quad (14)$$

and  $\|W\|_2 \leq \|W\|_F$ , so the Frobenius-product certificate  $U_F(w)$  is a conservative upper bound on the spectral-product one. The implementation constrains  $U_F(w)$  because its squared form

$$\prod_{l=1}^{D+1} \|W_l\|_F^2 \leq L^2 \quad (15)$$

is a smooth polynomial in the weights, convenient for the CasADi/IPOPT formulation; the trade-off is a potentially looser certificate than the spectral-norm product. For the scalar-input, scalar-output layers  $W_1 \in \mathbb{R}^{H \times 1}$  and  $W_{D+1} \in \mathbb{R}^{1 \times H}$  the spectral, Frobenius and Euclidean norms coincide; the distinction matters only for the internal  $H \times H$  matrices.

The depth experiment stacks 1 to 6 hidden layers of width 8 ( $n$  from 25 to 385), with  $\sigma = 0.05$ , seed 0,  $L = 4$ ,  $B = 6$  and 60 training / 40 test points. IPOPT satisfies the Frobenius-product certificate  $U_F(w)$  to solver tolerance at every depth, whereas unconstrained Adam's  $U_F(w)$  grows from about 1.75 at one layer to about  $2.5 \times 10^5$  at six (Fig. 6). Test errors stay near 0.003, so depth buys no accuracy, while the solve time grows from about 0.38 s to about 57 s for IPOPT and from about 0.27 s to about 21 s for penalty-Adam (Table III, Fig. 7). IPOPT converged for every configuration in the tested range of one to six hidden layers; the limit is expense, not failure.

## VI. CONCLUSION AND CRITICAL DISCUSSION

The true Lipschitz requirement is replaced conservatively by the computable certificate  $U(w)$ . When the resulting NLP is solved with IPOPT through CasADi, the network satisfies the certificate bound and norm ball to solver tolerance

TABLE III

DEPTH STUDY (WIDTH 8 PER LAYER, 1–6 HIDDEN LAYERS,  $n = 25$  TO 385): MATCHED IPOPT VS PENALTY-ADAM. DEPTH BUYS NO ACCURACY; IPOPT CONVERGES AT EVERY DEPTH BUT ITS COST GROWS STEEPLY.

| Depth | $n$ | IPOPT (hard) |          | penalty-Adam |          |
|-------|-----|--------------|----------|--------------|----------|
|       |     | Test         | Time [s] | Test         | Time [s] |
| 1     | 25  | 0.0031       | 0.38     | 0.0029       | 0.27     |
| 2     | 97  | 0.0033       | 1.33     | 0.0029       | 0.70     |
| 3     | 169 | 0.0031       | 7.81     | 0.0030       | 3.19     |
| 4     | 241 | 0.0031       | 15.59    | 0.0028       | 7.61     |
| 5     | 313 | 0.0030       | 31.15    | 0.0027       | 6.20     |
| 6     | 385 | 0.0032       | 57.44    | 0.0028       | 21.21    |

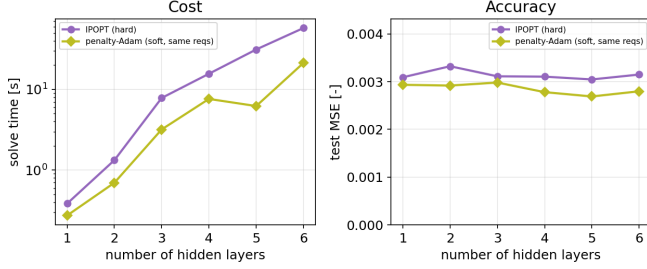


Fig. 7. Depth scaling over 1–6 hidden layers: solve time increases, especially for IPOPT, while test MSE remains nearly flat.

without a penalty weight. A soft-penalty Adam baseline must instead choose  $\rho$  and can fail to converge within a fixed iteration budget. Predictive errors are similar across the tested examples, and neither width nor depth improves accuracy on this small target; what grows is the cost of the hard solve, steeply with the number of weights. The approach is therefore suited to small constrained training problems where a certified requirement matters, rather than to large-scale neural-network training. Natural extensions are stochastic or mini-batch constraint handling for larger data, tighter spectral-norm certificates in place of the Frobenius product, and structure-exploiting KKT solvers to raise the size at which the hard solve stays affordable.

## REFERENCES

- [1] J. A. E. Andersson, J. Gillis, G. Horn, J. B. Rawlings, and M. Diehl, “CasADi: a software framework for nonlinear optimization and optimal control,” *Math. Program. Comput.*, vol. 11, no. 1, pp. 1–36, 2019.
- [2] A. Wächter and L. T. Biegler, “On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming,” *Math. Program.*, vol. 106, no. 1, pp. 25–57, 2006.
- [3] D. P. Kingma and J. Ba, “Adam: a method for stochastic optimization,” in *Int. Conf. Learn. Representations (ICLR)*, 2015.
- [4] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *Int. Conf. Learn. Representations (ICLR)*, 2019.
- [5] H. Gouk, E. Frank, B. Pfahringer, and M. J. Cree, “Regularisation of neural networks by enforcing Lipschitz continuity,” *Mach. Learn.*, vol. 110, no. 2, pp. 393–416, 2021.
- [6] T. Miyato, T. Kataoka, M. Koyama, and Y. Yoshida, “Spectral normalization for generative adversarial networks,” in *ICLR*, 2018.
- [7] J. Nocedal and S. J. Wright, *Numerical Optimization*, 2nd ed. New York: Springer, 2006.